

Table 1: Starting configuration for controlled comparisons. Experiment-specific changes are recorded before each comparison.

Parameter	Setting
Kafka deployment	Bitnami Kafka (KRaft) on Kubernetes
Brokers / topics / partitions	3 brokers; 1 fresh topic per run; 60 partitions
Replication factor	1
Producer / initial consumer pods	3 producers / 3 consumers
Consumer scaling	3 to 6 consumers when scaling is tested
Payload setting	100 bytes; actual message size recorded
Processing task	2,000 SHA-256 iterations per message; no added processing sleep
Reference input rate	700 messages/s in total across all three producers
Partition assignment	Cooperative Sticky for normal Kafka scaling; coordinated handover for targeted reassignment
Main run phases	1 minute warm-up, 10 minutes evaluation, 2 minutes drain (13 minutes total)
Scheduled intervention	60 seconds after evaluation begins; none in the baseline
Completion deadline	1 second from production to application completion, before offset-commit acknowledgment

Table 2: Completed experiments and planned comparisons. All rates are aggregate. Evaluation durations exclude one minute of warm-up and two minutes of drain. Planned settings and repetition counts will be fixed before testing.

Experiment	Workload and comparison	Trials / evaluation
Completed: 36 trials reported here		
Balanced calibration	600–1,500 msg/s; 60 balanced partitions; keep 3 consumers.	12; two per rate. 20 min (23 total).
Twelve hot partitions	700 msg/s; 80% to 12/60 partitions. Keep 3; redistribute 3; scale to 6 with targeted redistribution or Kafka rebalancing.	8; two per configuration. 10 min (13 total).
Four hot partitions	700 msg/s; 80% to 4/60 partitions. Redistribute 3 versus scale to 6 with targeted reassignment.	4; two per configuration. 10 min (13 total).
Earlier comparisons	700 msg/s; 80% to 12/60 partitions. Fixed ownership, redistribution and targeted scaling tests (appendix).	12 trials. 5 min (8 total).
Planned: not yet evaluated		
Hot-key splitting	Frequent-key input: splitting versus no action, scaling and redistribution.	To be specified.
Combinations and order	Splitting with scaling, redistribution or both; selected reverse action orders.	To be specified.
Automated policy	Heuristic controller versus HPA, KEDA and feasible research baselines.	To be specified.
Larger scale	Repeat selected comparisons with more partitions and consumers.	To be specified.
Applications	Drug–target interaction prediction and fire detection.	To be specified.

A Measurement Coverage

Table 3: Summary of the defined metrics, where their results are reported, and measurement limitations or remaining planned evaluations.

Metric family	Evidence in results	Interpretation
Partition lag and total lag	Main lag plots, diagnostic figure and lag-summary tables; older five-minute counterparts in the appendix.	Offsets, not guaranteed record counts.
Processing backlog and growth	Main backlog-growth plots and acknowledged-unfinished plots; growth is also retained for consumer-position lag.	Completion frontier and returned-record position have different definitions.
Maximum and mean partition lag	Bottom row of each diagnostic figure and sampled-peak tables.	Maximum and mean are taken across partitions at each valid observation.
Lag skew and window-mean skew	Runtime-state row shows window-mean skew; the summary table and separate diagnostic plot retain raw skew.	High relative skew can coexist with small absolute lag.
Hot set, persistence score and persistent-hot set	Persistent-hot row, detector settings and interpretation of early versus late observations.	Exploratory diagnostic, not a calibrated controller trigger.
Window-mean backlog and runtime state	Runtime-state figure and timestamped CSV/JSON, including persistent-hot partition IDs.	Reconstructed from retained observations; no adaptive controller was evaluated.
Completion mean, p50, p95 and p99	Outcome and latency-detail tables for each block.	Each quantile uses completed evaluation-born messages, not averaged rolling quantiles.
Processing-start latency and processing duration	Latency-detail tables and separation of queued delay from synthetic work.	Processing duration excludes evidence writing and commit bookkeeping.
Unfinished work and completion deadlines	Unfinished percentage and 1-second cohort deadline misses in main tables; 0.5-second sensitivity in appendix.	Overdue unfinished messages count as misses; they have no fabricated completion latency.
Observed-completion violation rate	Separate latency-detail column at 1 second.	Completion-attempt population during evaluation differs from the admitted cohort.
Input, useful throughput and attempt throughput	Main throughput plots and outcome tables; admitted rate and replay checks in text.	Attempts and distinct completions agree only when no duplicate completions occur.
Actual CPU and resident memory	Main process traces and observed resource integrals.	Consumer processes only; no full-cluster energy or monetary-cost claim.
Requested CPU and memory cost	Main resource tables, integrated over evaluation plus drain; detailed resource-cost figures retained with records.	Partial request coverage is unavailable for full-run comparison, not zero.
Interruption, transition backlog and recovery	Main intervention-cost tables, with operational endpoints identified.	Recovery is first threshold confirmation while input continues; not permanent stability.
Correctness and measurement quality	Identity reconciliation, handover validation, clock limitations and explicit coverage discussion.	Synthetic records do not prove exactly-once business effects at a durable external sink.
Controller overhead, splitting and state-transfer cost	Explicitly identified as unevaluated future measurements.	No adaptive controller, hot-key splitting or stateful transfer was evaluated in these trials.